The documentation and process conversion measures necessary to comply with this document shall be completed by 1 March 2021.

INCH-POUND

MIL-PRF-19500/678D
W/AMENDMENT 1
1 December 2020
SUPERSEDING
MIL-PRF-19500/678D
8 February 2020

PERFORMANCE SPECIFICATION SHEET

RECTIFIER, SEMICONDUCTOR DEVICE, SILICON, DUAL SCHOTTKY CENTER TAP, FOR POWER APPLICATIONS, SURFACE MOUNT, TYPES 1N6840 AND 1N6841, QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a silicon, dual Schottky power rectifier in a surface mount package. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outlines. The device package outline is a TO-276AA (SMD-0.5) in accordance with figure 1 for all encapsulated device types.

1.3 Maximum ratings. Unless otherwise specified, T_C = +25 °C.

Type	V_{RWM}	I_O (1) T_C= +100 °C	I_{FSM} T_C =+25 °C t_p= 8.3 ms	C_J at 5 V	$R_{\theta JC}$ (2)	T_{STG} and T_J
	V	A dc	A (pk)	pF	°C/W	°C
1N6840U3	35	10	200	400	2.8	−65 to +150
1N6841U3	45	10	200	400	2.8	−65 to +150

(1) Derate linearly at 200 mA/°C from $T_J = T_C$ =+100 °C to + 150 °C.
(2) Each individual diode.

1.4 Primary electrical characteristics. $R_{\theta JC}$ = 1.7 °C/W maximum for entire package (both legs tied together).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet are "JAN", "JANTX", "JANTXV" and "JANS".

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.5.2 Device type. The designation system for the device types of power rectifiers covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The power rectifiers of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the power rectifiers covered by this specification sheet are as follows: "6840" and "6841".

1.5.3 Suffix symbols. The following suffix symbols are incorporated in the PIN for this specification sheet:

U3	Indicates a metal lid 3 pad surface mount package similar to a TO-276AA (SMD-0.5) (see figure 1).

1.5.4 Lead finish designator. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

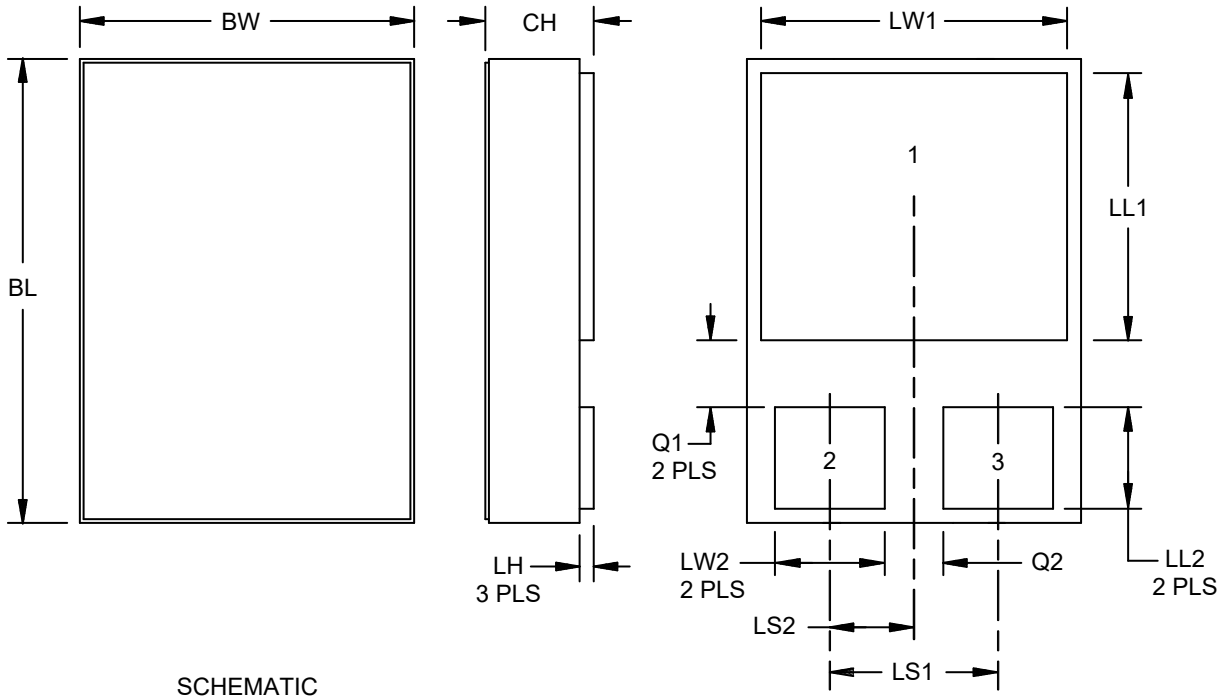
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions				Symbol	Dimensions			
	Inches		Millimeters			Inches		Millimeters	
	Min	Max	Min	Max		Min	Max	Min	Max
BL	.395	.405	10.03	10.29	LS1	.150 BSC		3.81 BSC	
BW	.291	.301	7.39	7.65	LS2	.075 BSC		1.91 BSC	
CH	.108	.124	2.74	3.15	LW1	.281	.291	7.14	7.39
LH	.010	.020	0.25	0.51	LW2	.090	.100	2.29	2.54
LL1	.220	.230	5.59	5.84	Q1	.030		0.76	
LL2	.115	.125	2.92	3.18	Q2	.030		0.76	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Terminal 1 is common cathode. Terminal 2 is anode 1. Terminal 3 is anode 2.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions and configuration of TO-276 (SMD-0.5) package.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as specified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and herein. The device package style is a TO-276AA in accordance with figure 1 herein.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2)

3.4.2 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I herein.

3.7 Workmanship. Power rectifiers shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E inspection. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (quality levels JANS, JANTXV and JANTX only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS level	JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.2).	Thermal impedance (see 4.3.2).
9 and 10	Not applicable.	Not applicable.
11	V_{F2} and I_{R1}.	V_{F2} and I_{R1}.
12	See 4.3.1, t = 240 hours.	See 4.3.1, t = 48 hours.
13	Subgroups 2 and 3 of table I herein; ΔV_{F2} = ±50 mV, ΔI_{R1} = 100 percent of initial value or 25 µA, whichever is greater.	Subgroup 2 of table I herein; ΔV_{F2} = ±50 mV, ΔI_{R1} = 100 percent of initial value or 25 µA, whichever is greater.
17	Dielectric withstanding voltage (see 4.3.3).	Dielectric withstanding voltage (see 4.3.3).

(1) Shall be performed anytime after temperature cycling, screen 3a. Quality levels JANTX and JANTXV do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in shall be performed in accordance with condition A of method 1038 of MIL-STD-750. The following details shall apply: T_C = + 125 °C; V_R = 80 percent of rated V_{RWM} dc.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_M, t_{SW}, t_{MD} (and V_H where appropriate). See table III, group E, subgroup 4 herein.

4.3.3 Dielectric withstanding voltage. The test for dielectric withstanding voltage shall be performed in accordance with method 1081 of MIL-STD-750. After the test, endpoints in accordance with subgroup 2 of table I herein shall be taken. The following details shall apply:

a. Magnitude of test voltage……600V dc.

b. Duration of application of test voltage……..15 seconds (min).

c. Points of application of test voltage……..All leads to case (bunch connection).

d. Method of connection……..Mechanical.

e. Kilovolt-ampere rating of high voltage source…….1,200 V/1.0 mA (min).

f. Maximum leakage current……..1.0 mA.

g. Voltage ramp up time.....500 V/second

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in 4.4.2.1 for quality level JANS and 4.4.2.2 for quality levels JAN, JANTX and JANTXV. Delta electrical measurements shall be in accordance with table II herein.

4.4.2.1 Quality level JANS (table E-VIA of MIL-PRF-19500).

Subgroup	Method	Condition
B3	4066	I_{FSM} = 200 A; 1 surge of 8.3 ms superimposed on I_O. Condition A, T_A = 25 °C; V_R = 0; I_O = 10 A continuous half-wave.
B4	1037	Condition for intermittent operation life for each diode is as follows: I_F or I_O = 2 A minimum; ΔT_J = 85 °C, + 15 °C, −5 °C.
B5	1038	Condition B, T_{JMAX} = +150 °C, T_A = 100 °C, 240 hrs.
B6	3101	Limit for thermal resistance is 2.80 °C/W per side.

4.4.2.2 Quality levels JAN, JANTX, and JANTXV (table E-VIB of MIL-PRF-19500).

Subgroup	Method	Condition
B2	4066	I_{FSM} = 200 A; 1 surge of 8.3 ms superimposed on I_O. Condition A, T_A = 25 °C; V_R = 0; I_O = 10A continuous half-wave.
B3	1037	Condition for intermittent operation life for each diode is as follows: I_F or I_O = 2 A minimum; ΔT_J = 85 °C, + 15 °C, −5 °C.
B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and herein. Delta electrical measurements shall be in accordance with table II herein.

Subgroup	Method	Condition
C3	2006	Conditions are as follows: X_1, Y_1, Z_1 and Z_2 axis.
C5	3101	See 4.3.2, $R_{\theta JC}$ = 2.80 °C/W maximum, per side.
C6	1037	Condition for intermittent operation life for each diode is as follows: I_F or I_O = 2 A minimum; ΔT_J = 85 °C, + 15 °C, −5 °C.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 3/	3101	See 4.3.2	$Z_{\theta JX}$			°C/W
Forward voltage	4011	Condition B. I_{FM} = 3 A (dc) pulsed (see 4.5.1) I_{FM} =10 A (dc) pulsed (see 4.5.1) I_{FM} =20 A (dc) pulsed (see 4.5.1)	 V_{F1} V_{F2} V_{F3}		 0.62 0.75 0.88	 V dc V dc V dc
Reverse current leakage 1N6840U3 1N6841U3	4016	Condition A or B, pulsed (see 4.5.1) V_R = 35 V dc V_R = 45 V dc	I_{R1}		100	µA dc
Subgroup 3						
High temperature operation:		T_A = +100 °C				
Reverse current leakage 1N6840U3 1N6841U3	4016	Condition A or B, pulsed (see 4.5.1) V_R = 35 V dc V_R = 45 V dc	I_{R2}		15	mA dc
Forward voltage	4011	Condition B I_F = 10 A (dc) pulsed (see 4.5.1) I_F = 20 A (dc) pulsed (see 4.5.1)	 V_{F4} V_{F5}		 0.63 0.725	 V dc V dc
Low temperature operation:		T_A = −55 °C				
Forward voltage	4011	Condition B I_F = 10 A (dc), pulsed (see 4.5.1)	V_{F6}		0.85	V dc
Subgroup 4						
Junction capacitance	4001	V_R = 5 V dc; f = 1 MHz, V_{SIG} = 50 mV (p-p) (max)	C_J		400	pF
Subgroup 5						
Insulation resistance	1016	V_R = 600 V dc; from lid to bottom case. All terminals shorted.	I_{RES}		10	µA

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 6 and 7 Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ All measurements are for each side.
3/ If this test is performed 100 percent in screening, this test need not be performed in group A.

TABLE II. Delta electrical measurement for groups B, C and E. 1/ 2/ 3/ 4/ 5/

Step	Inspection	MIL-STD-750		Symbol	Limits		Units
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	Condition B, I_F = 10 A (pk) pulsed (see 4.5.1)	ΔV_{F2}	±50 mV dc from initial reading.		
2.	Reverse current leakage	4016	Condition A or B, $V_R = V_{RWM}$, pulsed (see 4.5.1)	ΔI_{R1}	±100 percent from initial reading or ±25µA, whichever is greater.		

1/ The delta electrical measurements for group B inspection for quality level JANS (table E-VIA of MIL-PRF-19500) are as follows:
   a. In addition to the measurements specified for subgroup 3, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 4, the measurements of steps 1 and 2 of this table shall also be taken.
   c. In addition to the measurements specified for subgroup 5, the measurements of steps 1 and 2 of this table shall also be taken.
2/ The delta electrical measurements for group B inspection for quality levels JAN, JANTX, JANTXV (table E-VIB of MIL-PRF-19500) are as follows:
   a. In addition to the measurements specified for subgroup 2, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 3, the measurements of steps 1 and 2 of this table shall also be taken.
   c. In addition to the measurements specified for subgroup 6, the measurements of steps 1 and 2 of this table shall also be taken.
3/ The delta electrical measurements for group C inspection (table E-VII of MIL-PRF-19500) are as follows:
   a. In addition to the measurements specified for subgroup 3, the measurements of steps 1 and 2 of this table shall also be taken for all quality levels.
   b. In addition to the measurements specified for subgroup 6, the measurements of steps 1 and 2 of this table shall also be taken for all quality levels.
4/ The delta electrical measurements taken during group E inspection (table E-IX of MIL-PRF-19500 and table III herein) for subgroups 1, 2 and 5 shall include the measurements of steps 1 and 2 of this table.
5/ Devices which exceed the table I limits for these tests shall not be accepted.

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TABLE III. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification inspection
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Thermal shock (temperature cycling)	1051	Condition G, 500 cycles.	
Hermetic seal	1071	Fine and gross leak.	
Electrical measurements		See table I, subgroup 2 and table II.	
Subgroup 2			45 devices, c = 0
Burn-in (steady-state blocking life)	1038	t = 1,000 hours, T_J = + 125 °C; V_R = 80 percent of rated V_{RWM}.	
Electrical measurements		See table I, subgroup 2 and table II.	
Subgroup 3			
Not applicable			
Subgroup 4			Sample size, N/A
Thermal impedance curves	3101	See MIL-PRF-19500.	
Subgroup 5			5 devices, c = 0
Surge	4066	Condition A, T_A = +25 °C I_{FSM} = 200 A for each leg, one surge of 8.3 ms superimposed on I_O. V_R = 0; I_O = 10 A (pk) half sine wave, continuous.	
Electrical measurements		See table I, subgroup 2 and table II.	

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

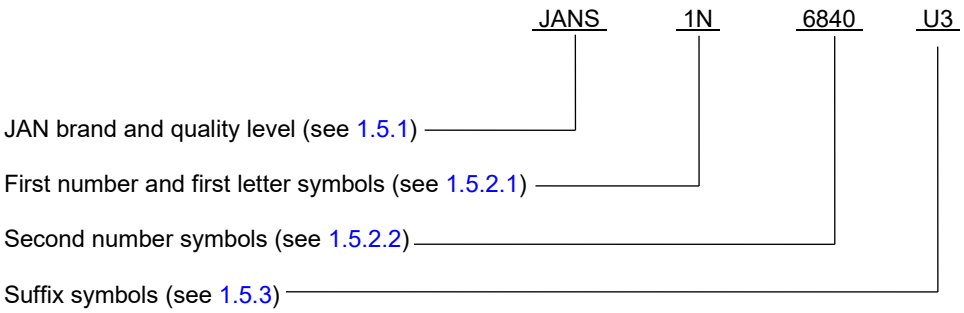
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form:



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6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

Base quality level	Quality level "TX"	Quality level "TXV"	Quality level "S"
JAN1N6840U3	JANTX1N6840U3	JANTXV1N6040U3	JANS1N6840U3
JAN1N6841U3	JANTX1N6841U3	JANTXV1N6841U3	JANS1N6841U3

6.6 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

* 6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
* Navy − SH
Air Force − 85
NASA − NA
DLA − CC

Review activity:
Air Force − 19

Preparing activity:
DLA − CC

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